

MICROCONTROLLER ARCHITECTURE AND PROGRAMMING

22EECF202/22EEIP213

COURSE PROJECT –MODULE DEVELOPMENT

Team: 05

Div: B

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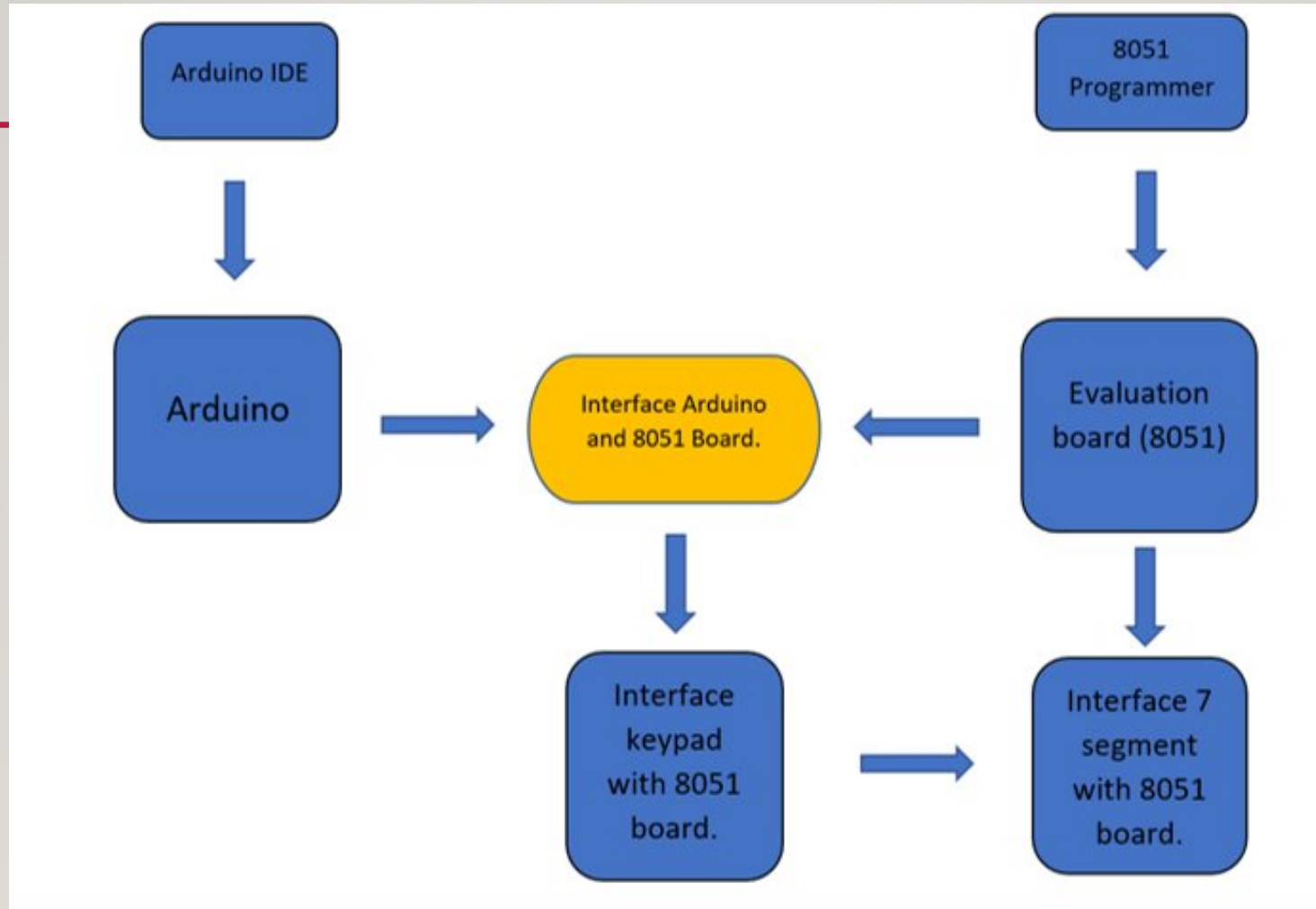
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PROBLEM STATEMENT

- KEYPAD INTERFACING-(MODULE DEVELOPMENT) USING 89S52 MICROCONTROLLER IC.

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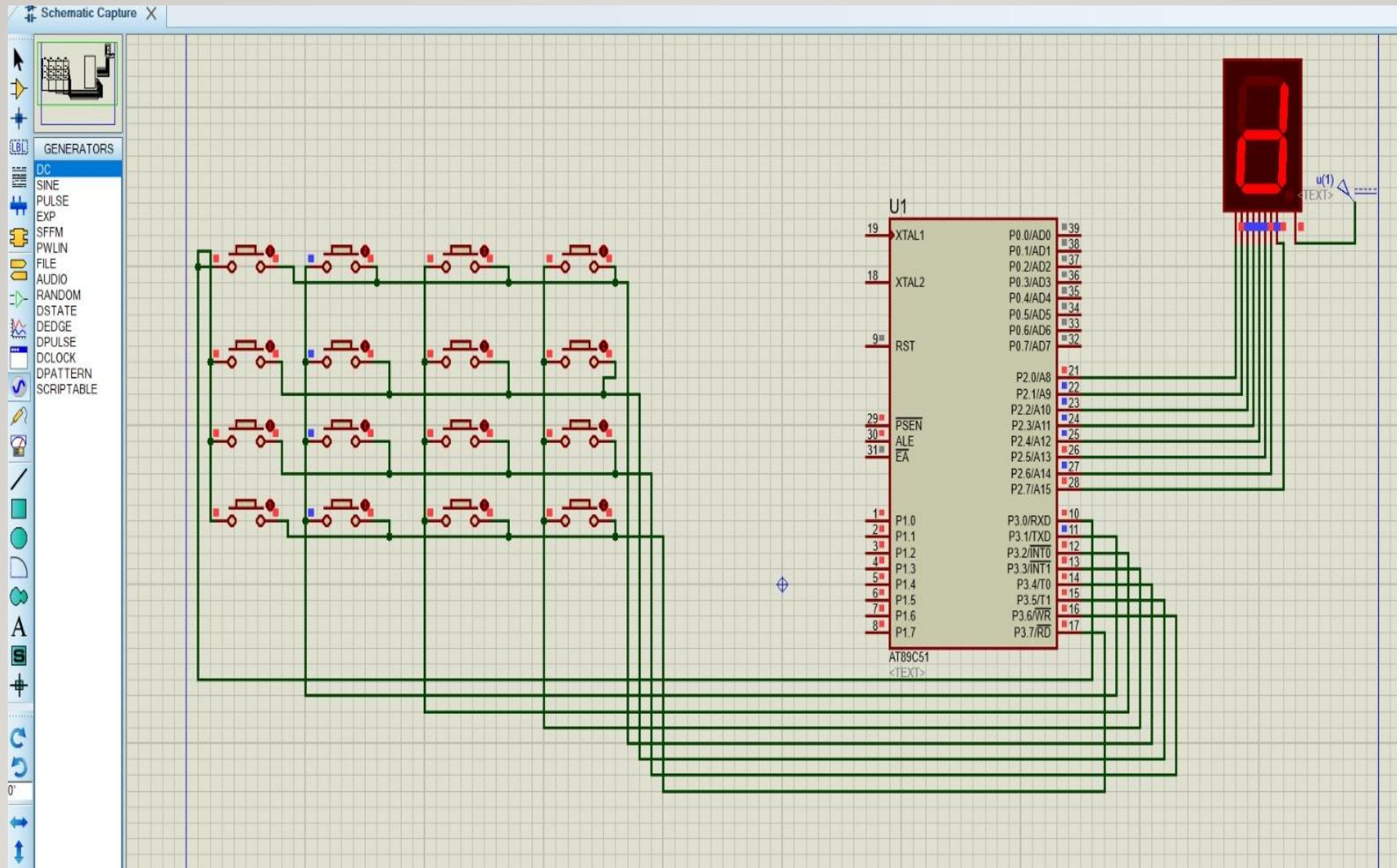
FUNCTIONAL BLOCK DIAGRAM



INTERFACING DIAGRAM

Microcontroller Architecture and Programming

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CODE

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Project Workspace	
Register	Value
[-] Regs	
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
[-] Sys	
a	0x00
b	0x00
sp	0x07
sp_max	0x07
PC \$	C:0x08C8
auxr1	0x00
+ dptr	0x0000
states	459
sec	0.00016691
+ psw	0x00

```
001 //keypad interfcaing
002 #include<reg51.h>
003 //initialse pins for rows and columns to port3
004 sbit R0=P3^0;
005 sbit R1=P3^1;
006 sbit R2=P3^2;
007 sbit R3=P3^3;
008 sbit C0=P3^4;
009 sbit C1=P3^5;
010 sbit C2=P3^6;
011 sbit C3=P3^7;
012 //FUNCTION PROTOTYPE
013 void display(unsigned int);
014 //main function
015 void main()
016 {
017     while(1)
018     {
019         R0=R1=R2=R3=1; //initialise all rows to high
020         R0=0;          //R0 is activated
021         if(C0==0)       //IF C0 IS ACTIVATED DISPLAY OUTPUT
022             display(0xC0);
023     }
024     R0=R1=R2=R3=1;
```


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Project Workspace

Register	Value
Regs	
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
Sys	
a	0x00
b	0x00
sp	0x07
sp_max	0x07
PC \$	C:0x08C8
auxr1	0x00
dp1r	0x0000
states	459
sec	0.00016691
psw	0x00

```
025 R0=0;
026 if (C1==0)
027 display(0xF9);
028
029 R0=R1=R2=R3=1;
030 R0=0;
031 if (C2==0)
032 display(0xA4);
033
034 R0=R1=R2=R3=1;
035 R0=0;
036 if (C3==0)
037 display(0xB0);
038
039 R0=R1=R2=R3=1;
040 R1=0;
041 if (C0==0)
042 display(0x99);
043
044 R0=R1=R2=R3=1;
045 R1=0;
046 if (C1==0)
047 display(0x92);
048
```

CODE

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Project Workspace

Register	Value
[-] Regs	
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
[-] Sys	
a	0x00
b	0x00
sp	0x07
sp_max	0x07
PC \$	C:0x08C8
auxr1	0x00
+ dptr	0x0000
states	459
sec	0.00016691
+ psw	0x00



```
048
049 R0=R1=R2=R3=1;
050 R1=0;
051 if (C2==0)
052   display(0x82);
053
054 R0=R1=R2=R3=1;
055 R1=0;
056 if (C3==0)
057   display(0xF8);
058
059 R0=R1=R2=R3=1;
060 R2=0;
061 if (C0==0)
062   display(0x80);
063
064 R0=R1=R2=R3=1;
065 R2=0;
066 if (C1==0)
067   display(0x90);
068
➔ 069 R0=R1=R2=R3=1;
070 R2=0;
071 if (C2==0)
```


CODE

Microcontroller Architecture and Programming

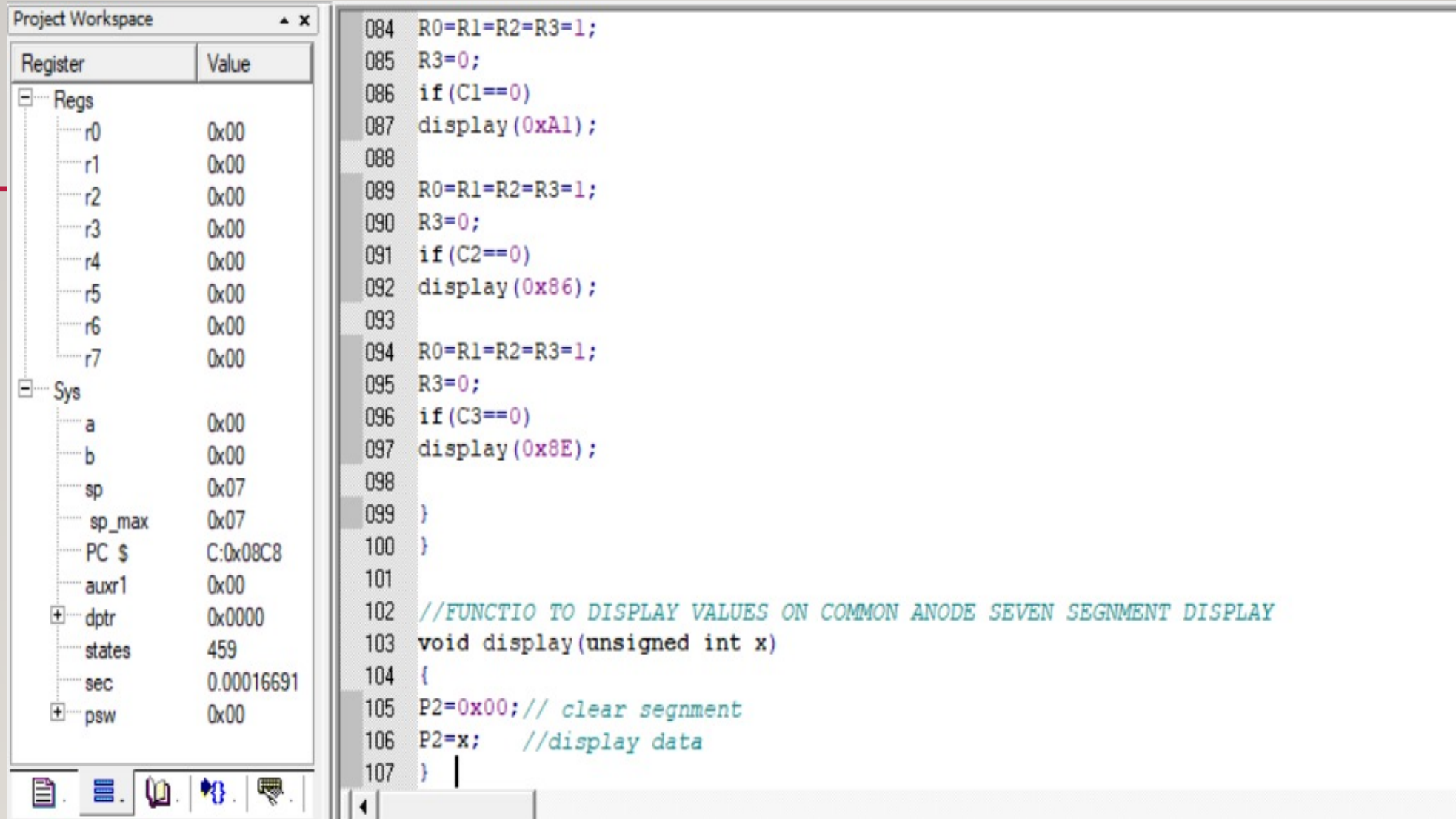
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Project Workspace

Register	Value
Regs	
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
Sys	
a	0x00
b	0x00
sp	0x07
sp_max	0x07
PC \$	C:0x08C8
auxr1	0x00
dptra	0x0000
states	459
sec	0.00016691
psw	0x00

```
071 if (C2==0)
072 display(0x88);
073
074 R0=R1=R2=R3=1;
075 R2=0;
076 if (C3==0)
077 display(0x83);
078
079 R0=R1=R2=R3=1;
080 R3=0;
081 if (C0==0)
082 display(0xC6);
083
084 R0=R1=R2=R3=1;
085 R3=0;
086 if (C1==0)
087 display(0xA1);
088
089 R0=R1=R2=R3=1;
090 R3=0;
091 if (C2==0)
092 display(0x86);
093
094 R0=R1=R2=R3=1;
```

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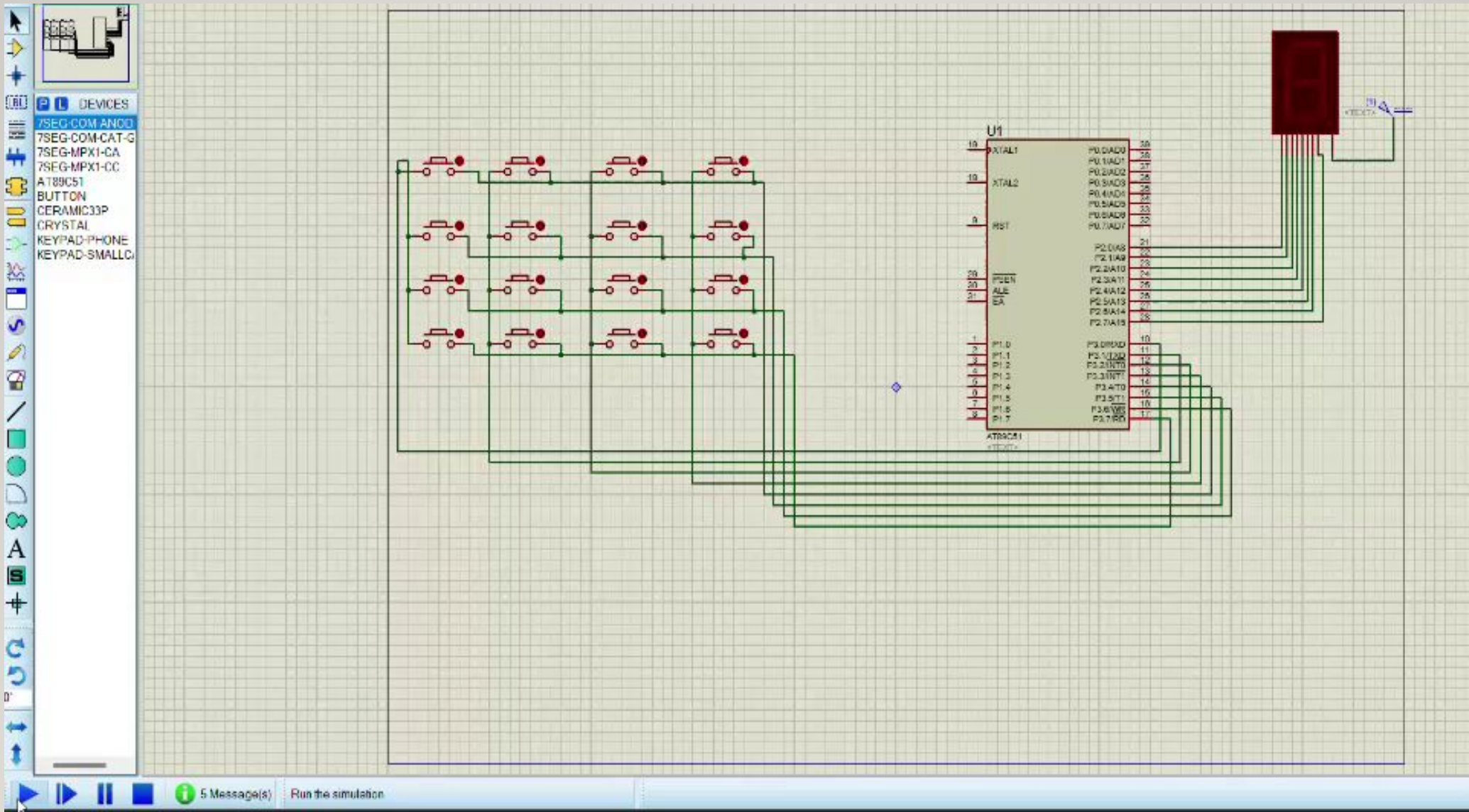
The screenshot displays a microcontroller development environment. On the left, the 'Project Workspace' window shows a tree view of registers and system variables. The 'Regs' section lists registers r0 through r7, all with a value of 0x00. The 'Sys' section lists system variables: a (0x00), b (0x00), sp (0x07), sp_max (0x07), PC \$ (C:0x08C8), auxr1 (0x00), dptr (0x0000), states (459), sec (0.00016691), and psw (0x00). On the right, the code editor shows assembly-like code with line numbers 084 through 107. The code includes conditional execution based on C1, C2, and C3, and a function definition for displaying values on a common anode seven-segment display.

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
a	0x00
b	0x00
sp	0x07
sp_max	0x07
PC \$	C:0x08C8
auxr1	0x00
dptr	0x0000
states	459
sec	0.00016691
psw	0x00

```
084 R0=R1=R2=R3=1;
085 R3=0;
086 if (C1==0)
087   display(0xA1);
088
089 R0=R1=R2=R3=1;
090 R3=0;
091 if (C2==0)
092   display(0x86);
093
094 R0=R1=R2=R3=1;
095 R3=0;
096 if (C3==0)
097   display(0x8E);
098
099 }
100 }
101
102 //FUNCTION TO DISPLAY VALUES ON COMMON ANODE SEVEN SEGMENT DISPLAY
103 void display(unsigned int x)
104 {
105   P2=0x00; // clear segment
106   P2=x;    //display data
107 }
```

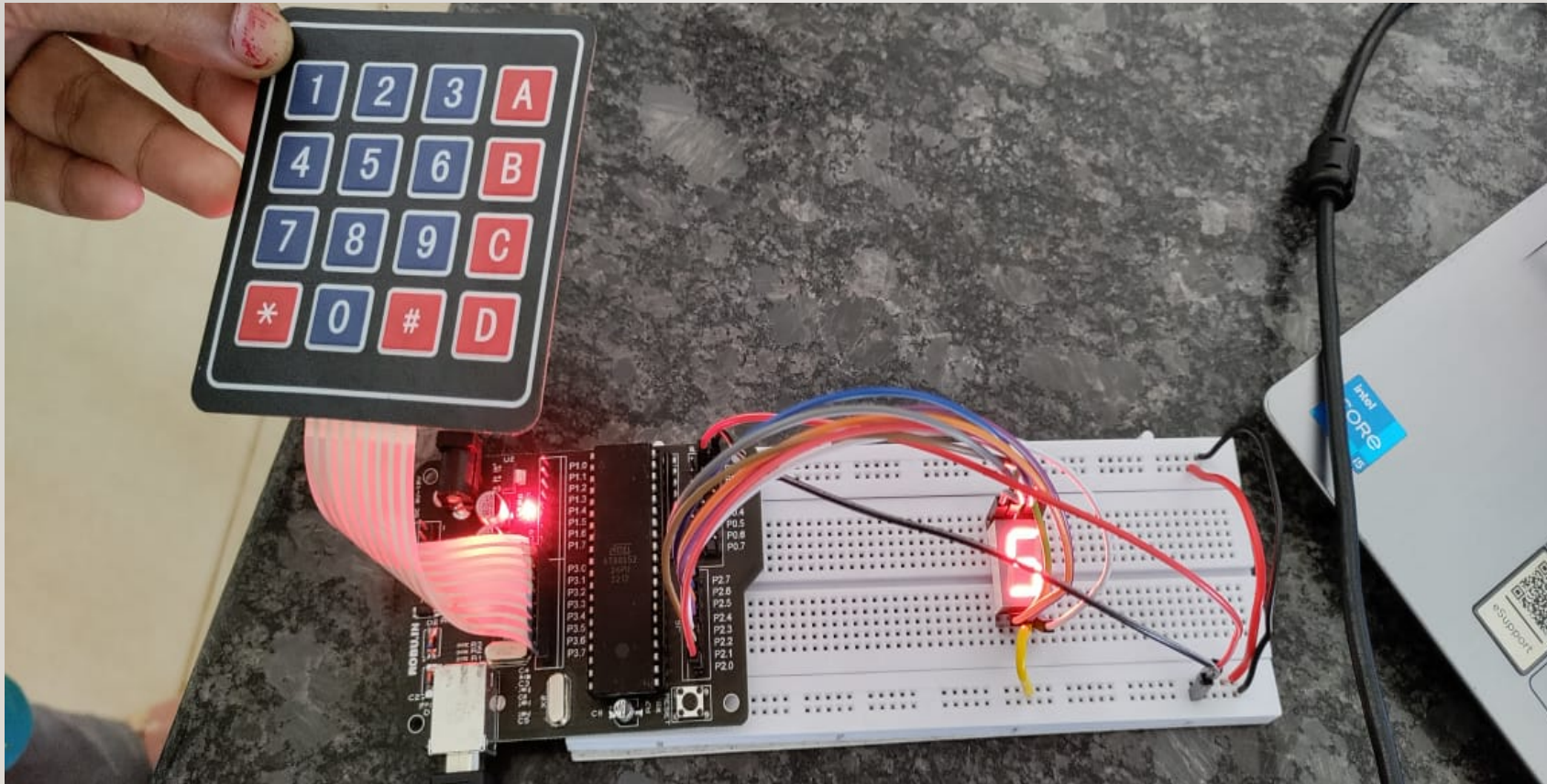

SIMULATION

Microcontroller Architecture and Programming



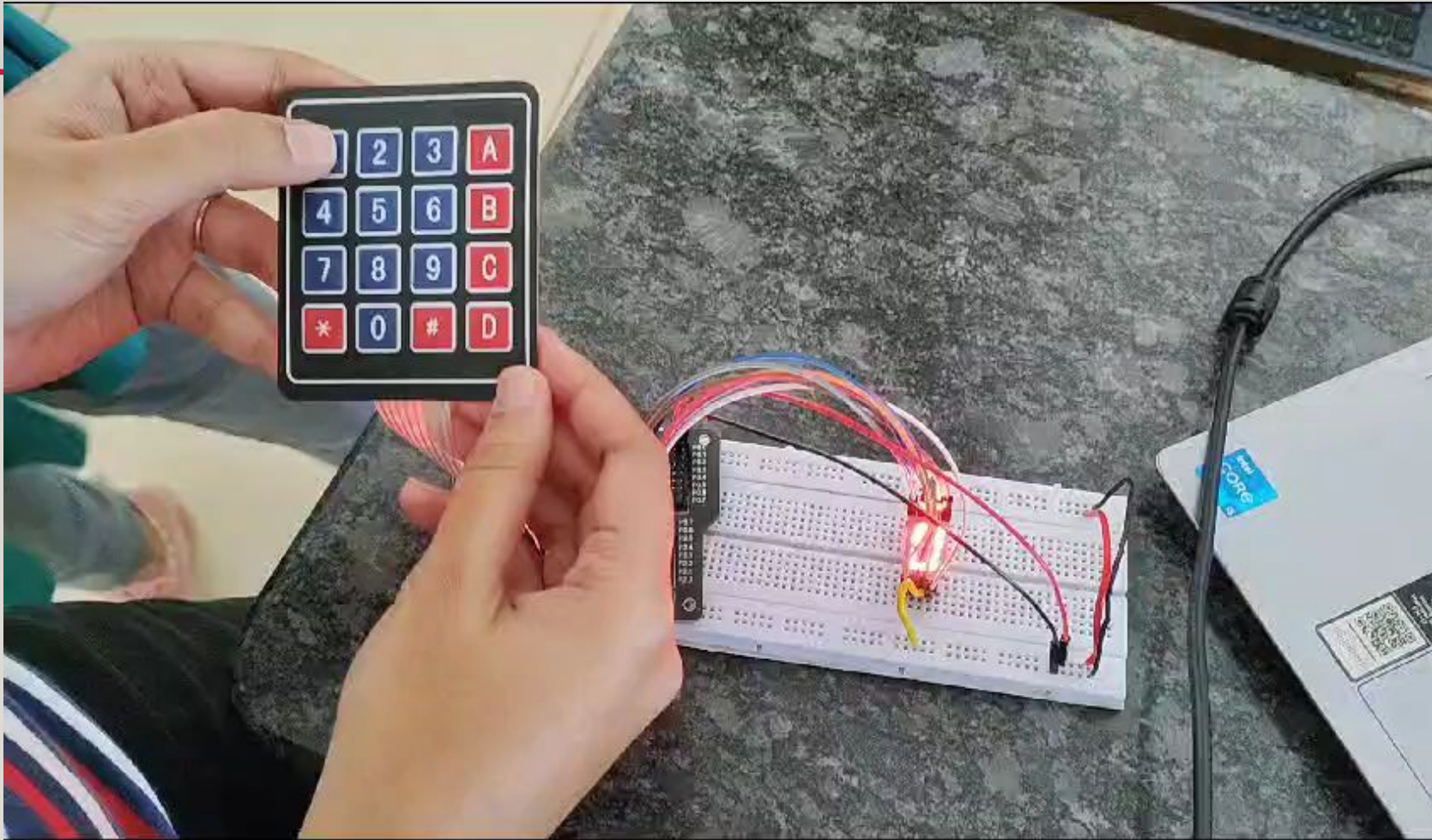
I2

HARDWARE RESULTS



I3

HARDWARE RESULTS



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LEARNING OUTCOMES

- Working with 89S52 microcontroller IC.
- Real life applications of Serial Communications.
- Working of a Keypad.
- How port pins of a microcontroller work as I/O pins.
- Programming a microcontroller with an Arduino UNO.
- Implementing a circuit on Printed Circuit Board.

Thank you

